MILITARY SPECIFICATION

SHELTER, ELECTRICAL EQUIPMENT S-336( )/TRC-112

1. SCOPE

1.1 Scope.- This specification covers the air or vehicular transportable shelter facility part of Radio Terminal Set AN/TRC-112( )(V) which is capable of transmitting and receiving a 12 or 24 channel pulse coded multiplexed base band transmission over troposcatter radio paths. (See 6.1 and 6.5).

1.2 Description.- Shelter, Electrical Equipment S-336( )/TRC-112(V) hereinafter called the shelter facility, includes a modified S-250( )/G Shelter and all racks, brackets, shelves, tie-downs, wave-guides, patch panel, other electrical components and wiring to enable installation and operation of the assemblage components of the AN/TRC-112( ). (See 6.1 and 6.2).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-P-116	Preservation, Methods of
MIL-C-3885	Cable Assemblies and Cord Assemblies (for use in Electronic and Associated Electrical Equipment)

FSC 5820

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MIL-W-6858	Welding, Resistance, Aluminum, Magnesium, Non-Hardening Steels or Alloys, Nickel Alloys, Heat-Resisting Alloys & Titanium, Spot & Seam
MIL-T-7928	Terminals, Lug & Splice, Crimp Style Copper
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finishes for Ground Signal Equipment
MIL-W-45205	Welding, Gas, Metal-Arc & Gas, Tungsten-Arc, Aluminum Alloys, Readily Weldable for Structures, Excluding Armor
MIL-S-55505	Shelter, Electrical Equipment (Containing Equipment) Packaging & Packing of

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures & Tables for Inspection by Attributes
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts
MIL-STD-252	Wired Equipment, Classification of Visual & Mechanical Defects
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-1235	Single and Multi-level Continuous Sampling Procedures & Tables for Inspection by Attributes

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DRAWINGS

ELECTRONICS COMMAND

SC-A-46691	RF Cable Assembly Tests
SC-D-602485	Schematic Diagram, Control Indicator Unit
DL-SC-B-630102	Shelter, Electrical Equipment S-336( )/TRC-112(V)
SC-D-630228	Power Schematic, AN/TRC-112( )(V)
SC-D-630302	Control and Signal Wiring Diagram, S-336( )/TRC-112(V)
SC-D-630303	Wave Guide and RF Wiring Diagram, S-336( )/TRC-112(V)

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies.)

3. REQUIREMENTS

3.1 Preproduction Sample.- Unless otherwise specified in the invitation for bids and contract, the contractor shall furnish one Preproduction Sample of the shelter facility for approval. (See 4.3).

3.2 Construction.- The shelter facility shall be constructed in accordance with Drawing List DL-SC-B-630102. (See 4.6).

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3.3 Electrical Installation.-

3.3.1 Power System Wiring and Components.- Installation and wiring of power wires and components (including cable assemblies, receptacles, switches, meters, circuit breakers, neon lamps, fluorescent lamps, incandescent lamps, lighting fixtures, distribution assembly system and wiring components constituting a four-wire (115/230 V AC, plus ground electrical power distribution system) shall be in accordance with the drawings. The completed in-place power distribution system shall meet the following requirements:

3.3.1.1 Continuity of Conductors.- Electrical continuity shall exist between each pin of the power entrance connectors and respective circuit terminations thereof when tested in accordance with 4.7.1 and 4.7.1.1.

NOTE: Test to meet the requirements of 3.3.1.1 in accordance with 4.7.1 shall be accomplished following completion of power system wiring and prior to mounting of AC power receptacles in the wireway.

3.3.1.2 Dielectric Breakdown Voltage.- The insulation of all power wires, cables, and components shall withstand a potential of 500 volts DC applied continuously for not less than 30 seconds when tested in accordance with 4.7.1 and 4.7.1.2.

3.3.1.3 Insulation Resistance.- The resistance of insulation between any two circuit conductors and also between each of the three ungrounded power-line conductors and shelter facility ground terminal shall not be less than 10 megohms when tested in accordance with 4.7.1 and 4.7.1.3.

3.3.2 Power Distribution and Operational Functions.- Main power branch distribution and circuit switching functions thereof shall meet the operational requirements as denoted by Drawing SC-D-630228 when tested in accordance with 4.7.2.

3.3.3 Signal (Audio) and Control Wiring and Components.- Installation and wiring of signal and control wires, cables and wiring components (including binding posts, connectors,

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plugs, jacks, receptacles, switches, terminal strips, terminal blocks, and wiring devices) shall be in accordance with the drawings. The completed in-place signal (audio) and control system installation shall meet the following requirements:

3.3.3.1 Continuity.- There shall be circuit continuity between end terminations of all audio signal and control conductors when tested in accordance with 4.7.3 and 4.7.3.1.

3.3.3.2 Dielectric Breakdown Voltage.- The insulation of all signal and control wires, cables and components shall withstand a potential of 600 volts DC applied continuously for not less than 5 seconds when tested in accordance with 4.7.3 and 4.7.3.2.

3.3.3.3 Insulation Resistance.- The resistance of insulation between each signal and control conductor and all other conductors when connected to ground shall not be less than 100 megohms, (See 4.7.3 and 4.7.3.3).

3.3.4 Video and Radio Frequency (RF), System Cabling and Components.- Installation of video and RF cable assemblies, and connecting components (including plugs, connectors, adapters, fittings, receptacles, and lightning protectors) shall be in accordance with the drawings. The completed in-place video and RF system installation shall meet the following requirements:

3.3.4.1 Continuity of Conductors.- The point-to-point resistance between the following video and RF cable terminations shall be less than 1 ohm when tested in accordance with 4.7.4 and 4.7.4.1.

(a) Both end terminations of the inner conductor.

(b) Both end terminations of the outer conductor(shield).

3.3.4.2 Dielectric and Insulation Resistance Tests.- The insulation of all video and RF cables, components and lightning protectors shall withstand the following potentials with

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the insulation resistance not less than 100 megohms when tested in accordance with 4.7.4 and 4.7.4.2.

(a) 1500 volts DC applied continuously for not less than 30 seconds to all video cable assemblies terminated at the video entrance box with lightning protector elements removes.

(b) 1100 volts DC applied continuously for not less than 30 seconds to all video cable assemblies terminated at the video entrance box with lightning protector elements in place.

(c) 1700 volts RMS, 25-100 cycles, applied continuously for not less than 30 seconds to all RF transmission lines.

(d) 1500 volts DC applied continuously for not less than 5 seconds to all video cable assemblies terminated at the video patch panel.

3.3.4.3 Voltage Standing Wave Ratio (VSWR), RF Transmission Lines.- The VSWR of all completed RF transmission lines shall not exceed 1.20 at transmission frequencies of 4400 MHz and 5000 MHz when tested in accordance with 4.7.4 and 4.7.4.3.

3.3.5 Waveguide Transmission Assemblies and Components.- The installation of waveguides, waveguide switch, directional couplers, dummy load and fittings constituting the 4.4-5.0 GHz transmission system shall be in accordance with the drawings. The completed in-place installation shall meet the following requirements:

3.3.5.1 Voltage Standing Wave Ratio (VSWR), Waveguide Assemblies.- The VSWR of each complete in-place waveguide transmission assembly shall not exceed 1.25 at the frequencies of 4.4 and 5.0 GHz when tested in accordance with 4.7.5 and 4.7.5.1.

3.3.5.2 Power Input, Transmit Waveguide Assembly.- With the shelter waveguide port (outside) terminated in a suitable dummy load, the transmit waveguide assembly shall withstand an input power level of not less than 1,000 watts at the

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frequencies of 4.4 and 5.0 GHz without arcing. (See 4.7.5 and 4.7.5.2)

3.3.6 Control and Indicator Units Operational Functions.- Switching, control and indicating functions of the control and indicator unit assemblies shall meet the operational requirements as denoted by Drawing SC-D-602485 (See 4.7.6).

3.3.7 Cable Assemblies.- (See 4.4)

3.3.7.1 Cable Assemblies, Power and Audio Frequency.- All power and audio frequency cable and cord assemblies shall be fabricated in accordance with the drawings unless otherwise stated and shall meet the requirements and tests of MIL-C-3885.

3.3.7.2 Cable Assemblies, Radio Frequency.- Radio frequency cable assemblies required for operation of major items of equipment in 6.1 shall be fabricated in accordance with the drawings unless otherwise stated and shall meet the requirements and tests as indicated in Drawing SC-A-46691.

3.3.8 Suppression of Electromagnetic Interference.- The motors in the ventilating equipment and heaters used in the shelter facility shall meet the emanation limits of MIL-STD-461 for tactical equipment, if other than those items defined by the drawings (See 4.4).

3.4 System Operation.- With all operational equipments of the AN/TRC-112( ) assemblage installed, the operational requirements of all power, audio, control and radio frequency cable assemblies, switches, connectors, waveguide assemblies, control and indicator unit, and patch panel, shall be met when tested in accordance with 4.8.

3.5 Vehicular Transportation.- The shelter facility shall be capable of being transported over cross-country terrain by military vehicles without sustaining any permanent damage such as buckling, cracking, delamination, or other permanent deformation. (See 4.9).

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3.6 Railroad Transportation.- The shelter facility shall be capable of being loaded, blocked, and braced on a flatcar for shipment by rail. When so loaded, the shelter facility shall be subjected to and withstand the test of 4.10 without sustaining any damage as mentioned in 3.5.

3.7 Drops, Flat and Rotational.- The shelter facility shall be capable of withstanding drops of 18 inches onto concrete. The shelter facility shall be subjected to and withstand the tests of 4.11.1 and 4.11.2 without any damage as mentioned in 3.5.

3.8 Watertightness.- The shelter facility shall be capable of withstanding the test of 4.12 without any leakage into the facility.

3.9 Fording.- The shelter facility shall be capable of being immersed in water to a depth of 26 inches, measured from the bottom of the skids, for 1 hour without any leakage of water into the facility. (See 4.13).

3.10 Mounting Inserts.- All mounting inserts shall be installed without the use of any auxiliary materials such as adhesive, and when installed, each insert shall be capable of withstanding the tensile load and the torque load referenced in 4.14. There shall be no damage to the insert threads or heads, the insert shall not have rotated from its original position, and there shall be no permanent distortion of the shelter facility mounting surface or insert head or threads. The insert head (if any) shall be properly seated and tight against its bearing surface. Insert threads and outer face of the insert head, shall be free of paint.

3.11 Cleaning.- After fabrication, parts shall be cleaned in accordance with good commercial practice or as specified in the applicable portions of Specification MIL-F-14072. Cleaning processes shall have no deleterious effect. Corrosive material shall be removed completely before the parts are assembled. After assembly, units shall be cleaned thoroughly and shall be free from particles of solder, flux, scale, and any other foreign material. In addition, when necessary, such cleaning shall also be performed before and after final assembly of the units.

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3.12 Finish.- The shelter facility shall be finished in accordance with Specification MIL-F-14072 and the drawings. (See 4.4).

3.13 Marking.-

3.13.1 General.- Marking shall conform to Specification MIL-M-13231 and the drawings. (See 4.4).

3.13.2 Nameplate.- Each shelter facility shall be provided with a nameplate in accordance with the drawings. Each nameplate shall be imprinted with a sequential serial number.

3.13.3 Aircraft Loading Data Plate.- The shelter facility shall be provided with data plates in accordance with the drawings, imprinted with the dimensions which locate the centers-of-gravity of the shelter facility and the assemblage in the three reference planes; the gross weight of the shelter facility and assemblage; and the bearing pressure on the bottom of the skids. The shelter assemblage is the shelter facility with operational equipments or dummy loads in place. The locations of the centers-of-gravity shall be determined by the contractor using the preproduction sample.

3.13.4 Interior and Exterior Marking.- All interior and exterior marking of the shelter facility shall be as depicted on the drawings.

3.14 Riveting.- Riveted joints shall be tight, the joined parts shall be undamaged, and the rivet heads shall be properly seated and tight against their bearing surfaces.

3.15 Seal, Vapor.- All joints and raw edges of the framing for the power, signal, video, and antenna entrance boxes, air-conditioner, power amplifier air inlet and exhaust, and exhaust blowers shall be sealed to prevent entrance of moisture to the core material of the shelter panels and to the interior of the shelter. All joints shall be assembled and sealed in a manner to prevent collection and retention of moisture. All exterior rivets, screws, and fasteners shall be dipped in sealer prior to installation. The sealer shall be one part Silicone Construction Sealant SE1204 as made by General Electric Company or equal.

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3.16 Securing of Parts.- Brackets, lugs, flanges, inserts, bolts, and other mounting arrangements shall retain components of the facility securely so that loosening, wear of mounting arrangements or permanent separation of parts or components will not occur when the equipment is subjected to the specified drop and transport tests.

3.16.1 Screws, Threaded Devices, and Related Parts.- Screws, threaded devices and related parts shall be as defined by the drawings. Thread forming screws shall not be used.

3.17 Soldering.-

3.17.1 Solder and Flux.- Solder and flux shall be as defined on the drawings.

3.17.2 Acid or Acid Salts.- No acid or acid salts shall be used in preparation for or during soldering; however, exception is permitted for preliminary tinning of electrical connections and for tinning or soldering of mechanical joints not used to complete electrical circuits, but in no case shall acid or acid salts be used where they can come in contact with insulation material. Where acid or acid salts are used, as permitted above, they shall be completely neutralized and removed immediately after use.

3.17.3 Process.- There shall be no sharp points or rough surfaces resulting from insufficient heating. The solder shall feather out to a thin edge, indicating proper flowing and wetting actions, and shall not be crystallized, over heated or under heated. The minimum necessary amount of flux and solder shall be used for electrical connections. Any means employed to remove an unavoidable excess of flux shall not incur the risk of loose particles of flux, brush bristles, or other foreign material remaining in the equipment; flux being spread over a larger area; or damage to the equipment. Insulation material that has been subjected to heating during the soldering operation shall be undamaged and parts fastened thereto shall not have become loosened.

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3.18 Tropicalization of Material.- Canvas, webbing, wood, etc. shall be treated to be resistant to moisture and fungi as indicated on the drawings and as follows:

3.18.1 Treating Materials.- Treating materials containing a mercury-bearing fungicide shall not be used. The contractor shall determine that the treating material is compatible with the material or surface to be treated. Selection of treating materials shall be such that any increase in flammability of treated material will be held to the practical minimum.

3.18.2 Toxicity.- Treating materials shall cause no skin irritation or other injury to personnel handling the treated material during fabrication, transportation, operation, or maintenance of the equipment, or during use of the finished items when used for the purpose intended.

3.18.3 Flexibility.- Treatment shall not affect the flexibility of treated materials, to the extent that the equipment may fail to meet specified requirements when subjected to specified service conditions.

3.18.4 Statement of Treatment.- The contractor shall submit for approval, to the contracting officer, a statement describing in detail the materials to be treated and the treating materials and processes that he proposes to use. (See 6.2 (e)).

3.19 Welding.- Welded joints shall be as defined by the drawings and shall be such that grinding on the finish weld will be unnecessary except where grinding is specified on the drawings. Inert-gas-shielded arc welding (MIG) using helium, argon, or a mixture of the two shall be used for arc welding of aluminum. Welds shall conform to MIL-W-6858 Class B and MIL-W-45205 Class B as applicable; surfaces to be welded shall be cleaned in accordance with good commercial practice and shall be free of scale, paint, grease, and other foreign materials. Welds shall have thorough penetration and good fusion and shall be free of scabs, blisters, abnormal pock marks, cracks, voids, slag inclusions, and other harmful defects. Inert-gas-shielded arc welding shall be used for

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welding of aluminum except where spot welds are used. Welded assemblies shall be cleaned to remove any scale, oxidation products, and excess flux. Any acid used in cleaning shall be completely neutralized and removed. (See 4.4).

3.20 Wiring and Cabling.- Wiring and cabling shall be in accordance with the drawings and shall be neat and sturdy.

3.20.1 Slack.- Wires and cables shall be as short as practical except that sufficient slack shall be provided:

(a) To prevent undue stress on cable forms, wires, and connections.

(b) To facilitate field repair of broken or cut wires.

(c) To prevent chafing or breaking of wires due to repeated flexing of hinged parts.

3.20.2 Protection.- Wires and cables shall be so placed and protected as to avoid contact with rough or irregular surfaces or sharp edges. Wires shall not be bent sharply where they enter insulation material.

3.20.3 Splicing.- Wires in a continuous run between two terminals shall not be spliced during the wiring operation unless specifically indicated by the drawings.

3.20.4 Crimping of Solderless Terminal Lugs.- Crimping of solderless terminal lugs shall be so accomplished that the connections will meet the resistance (voltage drop) and tensile strength requirements and tests of Specification MIL-T-7928. (See 4.15).

3.20.5 Identification of Wiring.- Wiring shall be identified by color-coding and wire pair marking, as indicated on the drawings.

3.21 Interchangeability.- Like units, assemblies, subassemblies, and replaceable parts shall be physically and functionally interchangeable without modification of such

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items or of the equipment. (See 4.16). Individual items shall not be hand-picked for fit or performance. Reliance shall not be placed on any unspecified dimension, rating, characteristic, etc.

3.22 Workmanship.- The equipment shall be manufactured and assembled in accordance with the applicable portions of the following: (See 4.17).

3.2	Construction
3.3	Electrical installation
3.10	Mounting inserts
3.11	Cleaning
3.12	Finish
3.13	Marking
3.14	Riveting
3.15	Seal, Vapor
3.16	Securing of parts
3.17	Soldering
3.18	Tropicalization
3.19	Welding
3.20	Wiring and cabling
3.21	Interchangeability

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of Inspection. - Inspection shall be classified as follows:

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(a) Preproduction inspection (does not include preparation for delivery). (See 4.3)

(b) Inspection covered by subsidiary documents. (See 4.4)

(c) Quality conformance inspection.

(1) Quality conformance inspection of equipment before preparation for delivery. (See 4.5)

(2) Quality conformance inspection of preparation for delivery. (See 4.18)

4.3 Preproduction Inspection. - This inspection shall be performed by the contractor on the preproduction unit unless otherwise specified in the contract. (See 3.1) It shall consist of the preproduction inspection specified in the subsidiary documents covering the items listed in 4.4, and the inspection specified for Group A, Group B, and Group C (except EVR) (See Inspection TABLES I, II and III respectively). Preproduction inspection shall be performed in the following order, unless otherwise specified.

(a)	Mounting Inserts	(f)	Drop, Flat
(b)	Electrical Installation	(g)	Drop, Rotational
(c)	System Operation	(h)	Watertightness
(d)	Truck Transportability	(i)	Fording
(e)	Railroad Transportability		

The last tests performed shall be a repeat of the Electrical installation tests and System tests. Remaining preproduction tests specified for Groups A, B and C inspections but not included in the above list shall be performed in any order satisfactory to the Contracting Officer.

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4.4 Inspection Covered by Subsidiary Documents. - The following shall be inspected under the applicable subsidiary documents as part of preproduction inspection and inspection of equipment before preparation for delivery:

ITEM	WHERE REQUIRED
Cable Assemblies	3.3.7
Suppression, EMI	3.3.8
Finish	3.12
Marking	3.13
Welding	3.19
Solderless terminals	3.20.4

4.5 Quality Conformance Inspection of Equipment Before Preparation for Delivery. - The contractor shall perform the inspection specified in 4.4 and 4.5.1 through 4.5.4. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition, the Government - at its discretion - may perform all or any part of the specified inspection to verify the contractor's compliance with specified requirements. (See 6.6) Test equipment for Government verification inspection shall be made available by the contractor.

4.5.1 Group A Inspection. - Each unit of equipment shall be inspected for conformance to all the inspection and test requirements of TABLE I of this specification. In addition, equipment conforming to the above requirements shall be verified by a quality audit. The following procedure shall be used for the quality audit:

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TABLE I - GROUP A INSPECTION

INSPECTION	RQR PARA	INSP PARA	
Visual & Mechanical	3.22	4.17	
Electrical			
Power Wiring			
Continuity	3.3.1.1	4.7.1.1	
Dielectric Strength	3.3.1.2	4.7.1.2	
Insulation Resistance	3.3.1.3	4.7.1.3	ALL
Power Distribution	3.3.2	4.7.2.1 thru 4.7.2.2	DEFECTS ARE
Signal (Audio) & Control Wiring			MAJOR
Continuity & Circuit Resistance	3.3.3.1	4.7.3.1	
Dielectric Strength	3.3.3.2	4.7.3.2	
Insulation Resistance	3.3.3.3	4.7.3.3	
Video & RF Cabling			
Continuity & Circuit Resistance	3.3.4.1	4.7.4.1	
Dielectric Strength & Insulation Resistance	3.3.4.2	4.7.4.2	
Voltage Standing Wave Ratio	3.3.4.3	4.7.4.3	
Waveguide Assemblies			
Voltage Standing Wave Ratio	3.3.5.1	4.7.5.1	
Power Input, Transmit Waveguide	3.3.5.2	4.7.5.2	
Control & Indicator Functions	3.3.6	4.7.6	

NOTE: The order of inspection of Group A with respect to Groups B & C shall conform to the order specified in 4.3.

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At the start of this plan 100% of the units shall be inspected until five consecutive units are found free of defects. At that time inspection shall be reduced to one out of five of the units presented. Whenever a defective unit is found, the contractor shall immediately revert to 100% inspection, which shall then be continued until five consecutive units are found free of defects. At that time inspection shall again be reduced to one out of five of the units presented. Samples shall be selected on a random basis.

4.5.1.1 Nonconformance Group A.- A defect of unit of product is any nonconformance with the specified requirement. Any unit of product found defective during inspection shall be rejected. Rejected units may be repaired or corrected and then resubmitted for inspection. The contractor shall simultaneously notify the Procuring Contracting Officer, the government technical activity designated in the contract, and the quality assurance representative and include proposed corrective action whenever the following condition occurs: The contractor's Group A audit fails to clear for sampling, i.e. remains on 100% inspection, for 25 consecutive units. Notification is applicable only to units with major defects.

4.5.2 Group B inspection.- Equipment shall be assembled in discrete lots for quality conformance inspection, utilizing the procedures of MIL-STD-105 for lot formation. Each lot shall consist of units produced within a calendar month. This inspection, including sampling, shall conform to Table II and to the procedures for small-sample inspection of Standard MIL-STD-105, using the special inspection levels. Group B inspection shall be performed on units of production that have passed Group A inspection except for Mounting Inserts and Interchangeability Inspection which are performed prior. The government reserves the right to require complete retest of any sample which is reworked after failing Group B inspection.

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TABLE II - GROUP B INSPECTION

INSPECTION	REQR PARA	INSP PARA	AQL % Def (see 4.5.2.1)
Mounting inserts	3.10	4.14.2	4.0 (AOQL-5.0)
Interchangeability	3.21	4.16	4.0 (AOQL-5.0)
Watertightness	3.8	4.12	2.5 (AOQL-3.0)
Fording	3.9	4.13	2.5 (AOQL-3.0)

NOTE: The order of inspection of Group B with respect to Groups A & C shall conform to the order specified in 4.3. The order of inspection within Group B shall be as in Table II.

4.5.2.1 Group B sampling plans.- The Group B sampling plans, for the AQL's listed in Table II shall be as follows:

AQL	Inspection level
4.0%	S-4
2.5%	S-4

4.5.2.2 Alternate Group B inspection procedures.- Contractor may elect to use a continuous sampling and inspection plan instead of the lot inspection plan stated above. If this alternate is selected the procedure of implementation must be in accordance with MIL-STD-1235, "Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes" and the Group B sampling plans for the AOQL's listed in Table II of this specification shall be selected from Tables VIII, IX and X of MIL-STD-1235 which are for f=½. The number of units in the Production Interval shall be the entire production quantity unless otherwise specified in the contract. Before starting production Contractor shall notify the contracting officer in writing which method he intends to use. The method may not be changed during production without written approval of the government.

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4.5.3 Group C Inspection.- This inspection shall be as listed in Table III.

TABLE III - GROUP C INSPECTION

INPSECTION	RQR PARA	INSP PARA
Solderless-Terminal-Lug-Connection	3.20.4	4.15
Construction Verification	3.2	4.6
Mounting Inserts	3.10	4.14.1
System Operation	3.4	4.8
Truck Transportability	3.5	4.9
Railroad Transportability	3.6	4.10
Drops, Flat & Rotational	3.7	4.11

NOTE: The order of inspection of Group C with respect to Groups A & B shall conform to the order specified in 4.3 The order of inspection within Group C for all inspections listed shall be as listed in Table III. Units subjected to Group C Drop Tests shall subsequently be subjected to and meet the Watertightness & Fording tests of 4.12 & 4.13.

4.5.3.1 Sampling for Inspection of Construction Evaluation.- The Equipment Verification Review, in accordance with 4.6 shall be performed once on each part and assembly, randomly selected, and on their installation into one of first 10 shelter facilities produced. This inspection shall not be performed on the pre-production unit. This inspection shall be performed as the selected shelter is being assembled.

4.5.3.2 Sampling for Inspection of Solderless-Terminal-Lug-Connections.-Five specimens of each combination of wire and terminal produced each week shall be prepared for inspection. A specimen is a 1 foot length of wire on which a terminal lug has been applied, using the same tools and procedures used in crimping the terminal lugs in production.

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4.5.3.3 Sampling for Inspection of Drop Tests.- One sample, representative of the first 50 units or fraction thereof produced, shall be subjected to the drop tests. This sample shall be the same unit previously subjected to the truck and railroad transportation tests of 4.5.3.4. Thereafter, one sample from each additional 150 units or fraction thereof produced shall be subjected to the drop tests.

4.5.3.4 Sampling for other inspection.- For inspection tests listed below, one unit shall be selected at random from the first 10 units or fraction thereof which have been produced, unless otherwise specified in the contract. Selection of the unit will be made by the Government.

(a) Mounting inserts.
(b) System operation.
(c) Truck transportability.
(d) Railroad transportability.

4.5.3.5 Noncompliance.- All quality conformance inspections shall be halted, including Group A and Group B inspections, upon the occurrence of any Group C failure. The contractor shall immediately report in writing each Group C failure occurrence, including details of the failure and characteristics affected. The contractor shall immediately investigate the cause of failure and further report the results of investigation and details of the proposed corrective action on the processes and materials, as applicable, and on all units of product which are manufactured under the same conditions and which the Government considers subject to the same failure. Reports shall be forwarded to the responsible technical activity designated in the contract with a copy to the Government Procurement Quality Assurance Representative. After corrective action has been taken, additional sample units shall be subjected to Group C inspection (all inspections which the sample failed, at the option of the technical activity) and Group A and Group B inspections may be reinstituted. However, final acceptance and shipment will be withheld until Group C reinspection results have shown that the corrective action was effective and approved by the technical activity. (See "Delivery & Conditional Acceptance of Contract Items Subject to Group C Inspection" provisions elsewhere in the contract.)

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4.5.4 Reinspection of Conforming Group B and Group C Sample Units.- Unless otherwise specified, sample units which have been subjected to and pass Group B or Group C inspection, or both, may be accepted on contract, provided that they are resubjected to and pass Group A inspection after repair of all visible damage. (See 6.4)

4.6 Construction Verification.- The contractor shall perform an Equipment Verification Review (EVR), consisting of a complete technical audit of the equipment on order against the drawings (See 3.2) cited in the Technical Data Package (as supplemented by all approved engineering changes). EVR shall consist of the following:

(a) An audit to establish that "as-built" shelter facilities, including individual parts, their use in sub-assemblies, and assemblies (components), and their final installation into the shelter forming the shelter facility, are in accordance with the end product drawings.

(b) An audit to establish that "as-built" shelter facilities, including parts, components and their installation, meet the acceptance requirements specified in the Technical Data Package for each shelter part, components, and shelter facility. The audit shall record all acceptance test methods used together with resulting verification test data.

(c) Preparation of EVR report(s) containing the following information:

(1) Identification of parts, components, shelter facilities and details proving compliance with 4.6(a) and (b) above.

(2) Discrepancies noted.

(3) Corrective action taken.

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(d) Two (2) copies of the EVR report shall be submitted no later than 15 days prior to the submission of the first production unit or lot to the Government technical organization cited in the contract. The results of the EVR shall be subject to verification by authorized Government personnel at the time the first unit or production lot is offered for acceptance. Government verification will be to the depth and extent necessary to demonstrate that the "as-built" hardware is in accordance with the cited drawings. Copies of the EVR reports shall be made available to Government personnel during the verification. The contractor shall provide the following:

(1) Segregation of parts, components of shelter facilities, to permit reviewing Government personnel access for detailed inspection. The contractor shall also provide personnel to disassemble any parts of the shelter facility necessary to permit Government review of the shelter facility, its components and their installation, and to reassemble parts after inspection.

(2) Have as required, responsible personnel from each functional department available for discussions in their respective areas.

(3) Adequate administrative support for the EVR.

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4.7 Electrical Installation Tests. - The tests of 4.7.1 thru 4.7.6 shall be performed on each shelter facility. Before applying power to the shelter facility, an electrical ground at the power entrance box shall be made. When required by the test, power to be supplied shall be 115/230 volts AC, 60 cycles, single phase and 24 volts DC.

4.7.1 Power System Wiring and Components Installation Tests.- Prior to conducting power wiring and components tests of 4.7.1.1, 4.7.1.2 and 4.7.1.3, the following shall be accomplished: (Refer to Drawing SC-D-630228)

(a) Remove or disconnect the following electrical loads:

(1) All fluorescent, incandescent, and neon lamps (DS-1 thru DS-25).

(2) All fluorescent lamp starters.

(3) Exhaust air blowers.

(4) Control/Indicator unit power plug, P-20.

(b) Remove power distribution box cover panel and disconnect both conductors from the voltmeter terminals. Position or isolate disconnected leads so that they do not short to each other or to ground.

(c) Place the following circuit breakers and circuit controls in the ON position:

(1) Fluorescent lamp switch, S5.

(2) Incandescent cold start switch, S6.

(3) Blackout bypass switch, S4.

(4) Blower switches, S7 and S8.

(5) Main circuit breaker, CB-1

(6) All branch circuit breakers, CB-2 thru CB-14.

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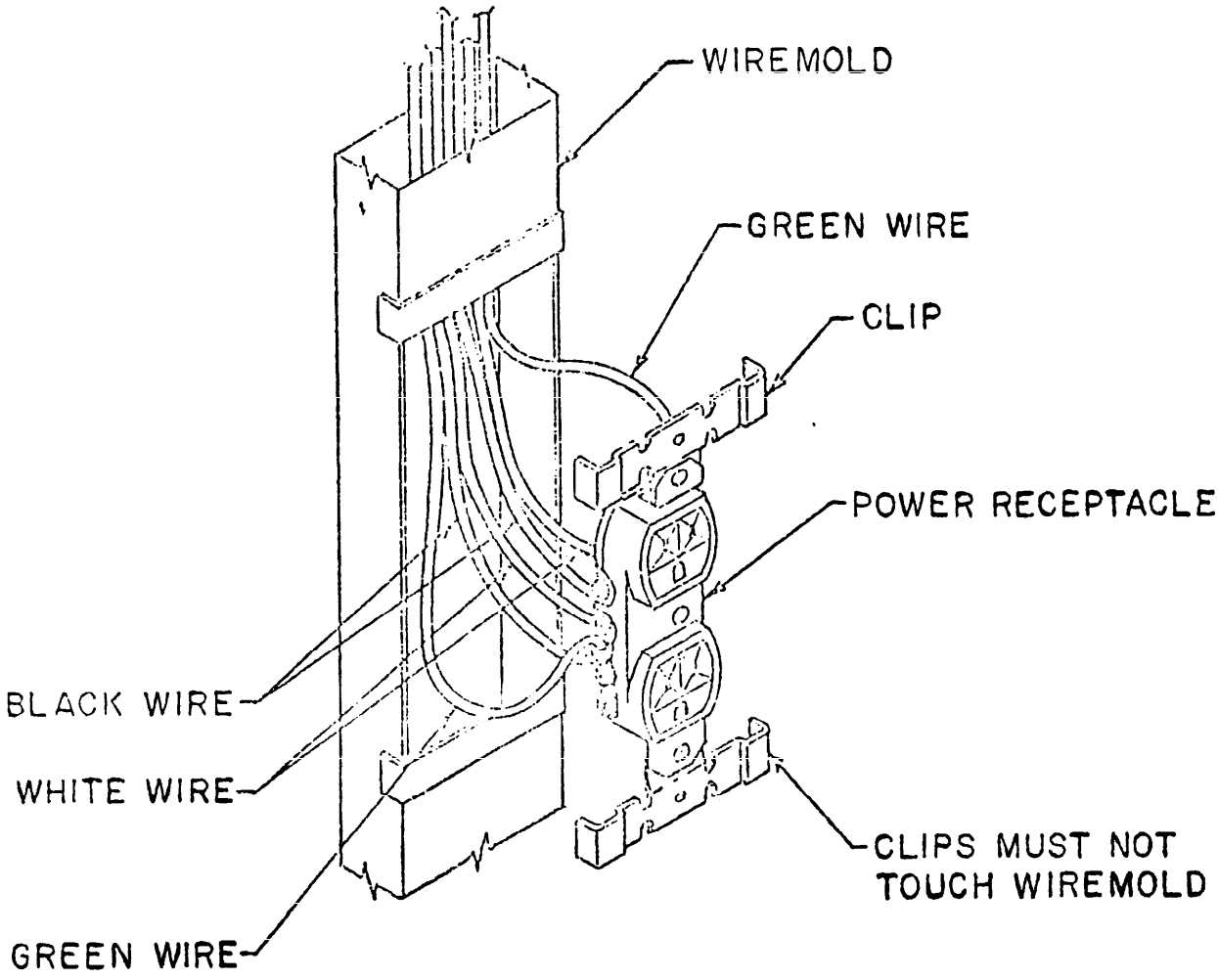
4.7.1.1 Continuity of Conductors. - Continuity tests shall be performed on each power-line circuit in accordance with the following paragraphs. (See 3.3.1.1) A low resistance ohmmeter or equivalent tester shall be used in performing the test.

(a) Remove and extend all connected power receptacles from the wireway so that the outlet mounting brackets do not make electrical contact to the wireway. (See Figure 1)

(b) Measure continuity as follows:

	FROM	TO
(1)	Pin 1 of J17	Grounding Contacts of Receptacles, J1 thru J15.
(2)	Pin 1 of J17	Ground Rod Terminal, E1.
(3)	Pin 1 of J17	Pin F of Plug, P17.
(4)	Pin 1 of J17	Pin D of Plug, P20.
(5)	Pin 1 of J17	Grounding Terminal, E2 in Power Distribution Box (Load Center).
(6)	Pin 2 of J17	Brass Contacts of Receptacles, J1, J3, J9, J10, J11, J13 and J14.
(7)	Pin 2 of J17	Pin C of Plug, P17.
(8)	Pin 2 of J17	Pin A of Plug, P20.
(9)	Pin 2 of J17	Black Wire Terminations at Lamps, DS-1 thru DS-9.
(10)	Pin 3 of J17	Silver Contacts of Receptacles, J1 thru J15.
(11)	Pin 3 of J17	Pin B of Plug, P17.

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POWER RECEPTACLE REMOVED & EXTENDED FROM WIREMOLD

FIGURE NO. 1

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	FROM	TO
(12)	Pin 3 of J17	Pin C of Plug, P20.
(13)	Pin 3 of J17	White Wire Terminations at Lamps, DS-1 thru DS-10.
(14)	Pin 4 of J17	Brass Contacts of Receptacles, J2, J4, J5, J6, J7, J8, J12 and J15.
(15)	Pin 4 of J17	Pin A of Plug, P17.
(16)	Pin 4 of J17	Pin B of Plug, P20, and Black Wire Termination at Lamp, DS-10.

(c) Place the 24 volt DC Power Transfer Switch (S9) in the POWER SUPPLY position. Measure continuity from the "To AN/GRC-106" Black Jack to "To DC Converter" Black (-) Conductor. Repeat continuity test from the Red Jack to Red (+) conductor.

(d) Place the 24 volt DC Power Transfer Switch (S9) in the BATTERY position. Measure continuity from Pin B of 24 volt DC Receptacle, J16 to "To AN/GRC-106" Black (-) Jack. Repeat continuity test from Pin A of J16 to the Red (+) Jack.

4.7.1.2 Dielectric Strength of Insulation. - Dielectric strength of insulation tests shall be made on each 60 cycle power line circuit conductor, including the neutral and each 24 volt DC Conductor. (See 3.3.1.2) Method 301 of MIL-STD-202 applies. A test voltage of 500 volts DC shall be applied for 30 seconds to the conductor under test with all other conductors placed at ground potential. Dielectric tests shall be made by applying the test voltage between ground and terminals of 120/240 volts AC Power Entrance Receptacle, J17 and 24 volts DC Receptacle, J16 as follows:

(a) Pin No. 2 of J17 (Pins No. 3 and No. 4 Grounded).

(b) Pin No. 3 of J17 (Pins No. 2 and No. 4 Grounded).

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(c) Pin No. 4 of J17 (Pins No. 2 and No. 3 Grounded)

(d) Pin No. A of J16 (Pin B Grounded)

(e) Pin No. B of J16 (Pin A Grounded)

4.7.1.3 Resistance of Insulation. - Insulation resistance tests shall be made on each power line conductor. (See 3.3.1.3) Method 302 of MIL-STD-202 applies. Test conditions, preparation, points of measurement and grounding of conductors shall be as given for 4.7.1.2 above. The insulation resistance of all conductors with respect to ground shall exceed 10 megohms.

4.7.2 Power Distribution and Operational Function Tests. - Power distribution and operational function tests shall be made subsequent to the tests of 4.7.1 and with all electrical sources and loads connected except as given by 4.7.2.(e). All components which are part of and/or associated with the electrical power system shall be in-place and connected. Prior to conducting the tests of 4.7.2.1 and 4.7.2.2, the following shall be accomplished (Refer to Drawing SC-D-630228):

(a) Secure blower exhaust port covers in OPEN position.

(b) Open shelter door.

(c) Place all circuit breakers (including main) on the 60 cycle 115/230 volts power distribution panel in the OFF position.

(d) Provide and place suitable test lamps or equivalent test indicators in each 60 cycle power outlet receptacle.

(e) Disconnect Plug, P20 from Control Indicator Unit.

(f) Provide and connect a suitable 120 volt test lamp or test indicator between Pins A and C and 240 volt test lamp or indicator between Pins A and B of Plug, P20.

(g) Provide and connect a suitable 120 volt test lamp or test indicator between Pins B and C and 240 volt test lamp or indicator between Pins A and C of Plug, P17.

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(h) Place all the lighting control switches in the ON position.

(i) Place the blackout by-pass switches in the BYPASS position.

(j) Set Thermo Switch, S10 to 170°F.

(k) Provide and connect a 24 volt DC meter to the "To AN/GRC-106" jacks (Positive Meter Lead to Red Jack; Negative Meter Lead to Black Jack).

4.7.2.1 60 Cycle AC Power Distribution Tests. -

(a) Apply 115/230 single phase 60 cycle power to connector in power entrance box. POWER ON Indicator Lamps, DS-9 and DS-10 shall light.

(b) Rotate Meter Switch, S2 to Positions L1, L2 and OFF. The Voltmeter, M-1 shall indicate specific line voltages and no voltage in the respective switched position.

(c) Place main circuit breaker (CB-1) in ON position. Turn each branch circuit breaker ON in sequence given below. The indication shall be as listed for each breaker.

CIRCUIT BREAKER NO.		INDICATION
(1)	CB-2	Both Test Lamps Connected to Plug, P17 Shall Light.
(2)	CB-3	Both Test Lamps Connected to Plug, P20 Shall Light.
(3)	CB-4	All Fluorescent and Incandescent Lamps (DS-1 thru DS-8) Shall Light.
(4)	CB-5	Test Lamps in Receptacles, J9, J10, J11 and J14 Shall Light.
(5)	CB-6	Blowers No. 1 and No. 2 Shall Operate and exhaust Air From the Shelter.

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CIRCUIT BREAKER NO.		INDICATION
(6)	CB-7	Test Lamp in Receptacle, J12 Shall Light.
(7)	CB-8	Test Lamp in Receptacle, J13 Shall Light.
(8)	CB-9	Test Lamp in Receptacle, J3 Shall Light.
(9)	CB-10	Test Lamp in Receptacle, J2 Shall Light.
(10)	CB-11	Test Lamps in Receptacles, J5, J6, J7 and J8 Shall Light.
(11)	CB-12	No Indication.
(12)	CB-13	No Indication.
(13)	CB-14	Test Lamp in Receptacle, J1 Shall Light.

(d) With all circuit breakers remaining in the ON position, operate power control switches to OFF position in sequence given below. The indication shall be as listed for each switch.

SWITCH NO.		INDICATION
(1)	S-5 (Fluorescent Lamps)	Fluorescent Lamps, DS-1 thru DS-4 Shall Extinguish.
(2)	S-6 (Cold Start)	Incandescent Lamps, DS-5 thru DS-8 Shall Extinguish.
(3)	S-7 (Blower-1)	Blower No. 1 Shall Stop.
(4)	S-8 (Blower-2)	Blower No. 2 Shall Stop.

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(e) With Blower No. 1 Switch (S-7) in the OFF position, place a jumper across the contacts of Thermo Switch, S10. Blower No. 1 shall operate. Remove jumper; blower shall stop.

Reset Thermo Switch, S10 to 85°F. Restore all Switches, S-5 thru S-8 to the ON position.

(f) With all circuit breakers and 60 cycle power control switches in the ON position, perform the following lighting blackout test:

(1) Place blackout by-pass Switch, S4 in the OFF position. Close the shelter door. All lamps (DS-1 thru DS-8) shall light.

(2) With the blackout by-pass Switch (S4) remaining in the OFF position, open the shelter door. All lamps (DS-1 thru DS-8) shall go out.

(g) Turn OFF the main and all branch line circuit breakers (CB-1 thru CB-13). Provide and connect a resistive load of 12 ohms (5 KW) between Terminals 1 and 2 of Terminal Block, TB-2. Turn ON main and branch circuit breakers, CB-1 and CB-3. Operate Meter Switch, S1 as given below. The indication shall be as listed.

METER SWITCH POSITION	INDICATION
(1) L1	Meter, M-2 shall read 20 amps ± 5 Percent.
(2) L2	Meter, M-2 shall read 20 amps ± 5 Percent.

4.7.2.2 24 Volt Power Distribution Tests. -

(a) Apply 24 volts DC to EXT Battery Receptacle, J16 with polarity as indicated (Negative to Pin B; Positive to Pin A).

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(b) Place 24 volt DC Transfer Switch (S9) in the EXT BATTERY position. Lamp, DS-25 shall indicate correct polarity and test meter (See 4.7.2(k))shall read 24 volts.

(c) Place 24 volts DC Transfer Switch (S9) in the DC CONVERTER position. Lamp, DS-25 shall go out and test meter shall read zero volts.

(d) With DC Transfer Switch (S9) remaining in the DC CONVERTER position, connect 24 volt DC Power to the DC Converter conductors (Positive to Red Conductor; Negative to Black Conductor). Lamp, DS-25 shall indicate correct polarity and test meter shall read 24 volts.

(e) Disconnect test meter and repeat test of 4.7.2.1(d) with reversed 24 volt DC polarity (Positive to Black Conductor; Negative to Red Conductor). Lamp, DS-25 shall indicate incorrect polarity.

4.7.3 Signal (Audio) and Control Wiring and Components Installation Tests. - Prior to conducting signal and control wiring and components tests of 4.7.3.1, 4.7.3.2 and 4.7.3.3, the main Power Circuit Breaker (CB-1) shall be placed in the OFF position and following removed or disconnected (Refer to Drawing SC-D-630302):

(a) Plugs, P-20 and P-21 to Control-Indicator Unit.

(b) Plug, P-22 to Waveguide Switch.

(c) Power Amplifier (AM-6090) Air Pre-Heating Element.

(d) 1100 OHM Resistor connected between Terminals 29 and 30 and condenser between terminals 17 and 19 of TB-1.

(e) Connect A jumper wire between terminals 17 and 19 of TB-1.

4.7.3.1 Continuity of Conductors. - Continuity tests shall be performed on each audio and control line circuit conductor, (See 3.3.3.1). A low resistance ohmmeter or equivalent continuity tester shall be used. Continuity measurements shall be made as follows:

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	FROM	TO
(a)	Pin A of P6	Pin B of P21
(b)	Pin C of P6	Pin A of P21
(c)	Pin F of P6	Pin E of P21
(d)	Pin M of P6	Pin G of P21
(e)	Pin N of P6	Pin H of P21
(f)	Pin P of P6	Pin D of P21
(g)	Pin R of P6	Pin C of P21
(h)	Pin S of P6	Blk Cond, CV-425(TB-1-29)
(i)	Pin T of P6	Wht Cond, CV-425 (TB-1-30)
(j)	Pin Z of P6	Shield, CV-425 PR (TB-1-31)
(k)	Pin K of P6	Pin A of P14
(l)	Pin L of P6	Pin B of P14
(m)	Pin U of P6	Pin D of P22
(n)	Pin V of P6	Pin F of P22
(o)	Pin H of P6	Pin B of P11
(p)	Pin J of P6	Pin A of P11
(q)	Pin W of P6	Pin C of P11
(r)	Pin X of P6	Pin K of P11
(s)	Pin Y of P6	Pin K of P11

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	FROM	TO
(t)	Pin A of J33	Pin A of P13
(u)	Pin B of J33	Pin B of P13
(v)	Pin C of J33	Pin C of P13
(w)	Pin D of J33	Pin D of P13
(x)	Pin E of J33	Pin E of P13
(y)	Pin F of J33	Blk CV-425 Cond (BP-1) and Binding Post No. 1
(z)	Pin G of J33	Wht CV-425 Cond (BP-2) and Binding Post No. 2
(aa)	Pin E of P20	Red Cond to P.A. Heater Element
(bb)	Pin F of P20	Blk Cond to P.A. Heater Element
(cc)	Pin G of P20	Pin C of P16
(dd)	Pin A of P16	TML 3 of TB-2
(ee)	Pin B of P16	Pin E of P16
(ff)	Pin F of P21	TML 13 of TB-1
(gg)	Pin M of P21	TML 14 of TB-1
(hh)	Pin P of P21	TML 8 of TB-1
(ii)	Pin N of P21	Pin D of P14
(jj)	Pin J of P21	Pin C of P22
(kk)	Pin K of P21	Pin A of P22
(ll)	Pin L of P21	Pin B of P22 and Pin E of P14

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	FROM	TO
(mm)	Pin A of P1	Pin A of J29
(nn)	Pin B of P1	Pin B of J29
(oo)	Pin E of P1	Pin E of J29
(pp)	Pin C of P1	Pin C of J29

4.7.3.2 Dielectric Strength of Insulation. - Dielectric Strength of Insulation Tests shall be made on each Signal (Audio) and Control Line Conductor (See 3.3.3.2). Method 301 of MIL-STD-202 applies. The test voltage shall be applied between the conductor(s) under test and Shelter Facility Ground Terminal E1 while all other signal and control conductors are placed at shelter ground by the following procedure:

(a) Using appropriate test jigs or fixtures, connect the following to Shelter Facility Ground Terminal (E1):

(1) All 19 conductor pin terminations of Plug, P6.

(2) All 7 conductor pin terminations of Receptacle, J33.

(3) Conductor Pin Terminations E, F and G of Plug, P20.

(4) Conductor Pin Terminations F, J, K, L, M, N and P of Plug, P21.

(5) Conductor Pin Terminations A, B, E and C of Plug, P1.

(b) Remove ground connection from Pin A of P6 and connect tester between this pin and Shelter Ground Terminal (E1).

(c) Adjust tester for 600 volt DC output and maintain this potential for a continuous period of not less than 5 seconds. The requirement of 3.3.3.2 shall be met. Disconnect tester and re-establish ground connection removed by 4.7.3.2(b) above.

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(d) Repeat 4.7.3.2b and 4.7.3.2c for each of the remaining conductor pin terminations (except pins Y and Z) of P6; each conductor pin terminations (except Pin C) of J33; conductor pin terminations E, F and G of P20; conductor pin terminations F, J, K, L, M, N and P of P21; and conductor pin terminations A, B, E and C of P1.

(e) Connect tester betweer Pin B of P16 and shelter ground. Repeat test of 4.7.3.2c. Disconnect tester.

4.7.3.3 Insulation Resistance.- Insulation resistance tests shall be made on each signal (audio) and control line conductor. Method 302 of MIL-STD-202 applies. The resistance shall be measured between the conductor(s) under test and shelter ground terminal (E1) while all other signal and control conductors are placed at shelter ground by the procedures of 4.7.3.2a, 4.7.3.2b and 4.7.3.3a below.

(a) Adjust tester to a voltage of not less than 200 nor more than 600 volts DC and maintain potential until resistance reading is steady or steadily increasing in value. The insulation resistance shall exceed 100 megohms. (See 3.3.3.3). Disconnect tester and re-establish ground connection removed by 4.7.3.2b.

(b) Repeat 4.7.3.2d.

(c) Repeat 4.7.3.2e.

(d) Remove temporary wire jumper between terminals 17 and 19 of TB-1 and reconnect condenser. Reconnect 110C OHM resistor (See 4.7.3(d) and 4.7.3(e).

4.7.4 Video and Radio Frequency (RF) Cabling and Components Tests.- (Refer to drawing no. SC-D-630303).

4.7.4.1 Continuity of Conductors.- Continuity tests which meet the requirements of 3.3.4.1 shall be performed on each video and RF coaxial line. A low resistance ohmmeter or equivalent instrument shall be used.

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(a) Continuity measurements shall be made as follows:

From	To
Pin of J30	Pin of J28
Shield of J30	Shield of J28
Pin of J31	Pin of J27
Shield of J31	Shield of J27
Pin of P2	Pin of J22
Shield of P2	Shield of J22
Pin of P3	Pin of J21
Shield of P3	Shield of J21
Pin of J32 (female)	Pin of J23
Shield of J32 (female)	Shield of J23
Pin of J32 (male)	Pin of J24
Shield of J32 (male)	Shield of J24
Pin of P9	Pin of P18
Shield of P9	Shield of P18
Pin of P8	Pin of P19
Shield of P8	Shield of P19
Pin of P5	Pin of P15
Shield of P5	Shield of P15
Pin of P10	Pin of P7
Shield of P10	Shield of P7

4.7.4.2 Dielectric and insulation resistance tests.- Dielectric and insulation resistance tests which meets the requirements of 3.3.4.2 shall be performed on each video and RF circuit in accordance with the following paragraphs. A non-destructive dielectric and insulation resistance tester shall be used in performing the test.

(a) At Cable Patch Panel make the following connections using Cable Patch Cords:

(1) From J23 to J21.

(2) From J24 to J22.

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(b) Remove all lightning protector elements at video entrance box.

(c) Apply test potential of 1500 volts DC for not less than 30 seconds between the following pin connections and terminal ground. The requirements of 3.3.4.2a and 3.3.4.2d shall be met.

(1) Pin of J32 (male).

(2) Pin of J32 (female).

(3) Pin of J30.

(4) Pin of J31.

(d) Replace lightning protector elements removed by step b above.

(e) Repeat (1) and (2) of step (c) above applying test potential of 1100 volts DC. The requirements of 3.3.4.2b shall be met.

(f) Connect tester between inner conductor of P5 and shell of plug.

(g) Adjust tester to 1700 RMS output and maintain this potential for a continuous period of not less than 30 seconds. The requirements of 3.3.4.2c shall be met.

(h) Repeat steps f and g above substituting P8.

(i) Repeat steps f and g substituting P9.

(j) Repeat steps f and g substituting P10.

4.7.4.3 Voltage Standing Wave Ratio (VSWR) Tests, RF Coaxial Transmission Lines.- VSWR tests shall be made on each in-place RF coaxial transmission line. The basic test set-up

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(2) Modulation transfer function (MTF) analyzer.

shall be as shown in Figure 2. Tuners and/or pads shall be included where necessary to reduce the residual VSWR of test set-up to a level which will assure accurate measurements. VSWR measurements shall be made as follows:

(a) Connect coaxial cable termination (HP909A or equal) to plug, P19 and slotted line to plug, P8.

(b) Measure VSWR at frequencies of 4.4 and 5.0 GHz. The requirements of 3.3.4.3 shall be met.

(c) Connect coaxial cable termination to plug, P18 and slotted line to plug, P9.

(d) Repeat 4.7.4.3b.

(e) Connect coaxial cable termination to plug, P15 and slotted line to plug, P5.

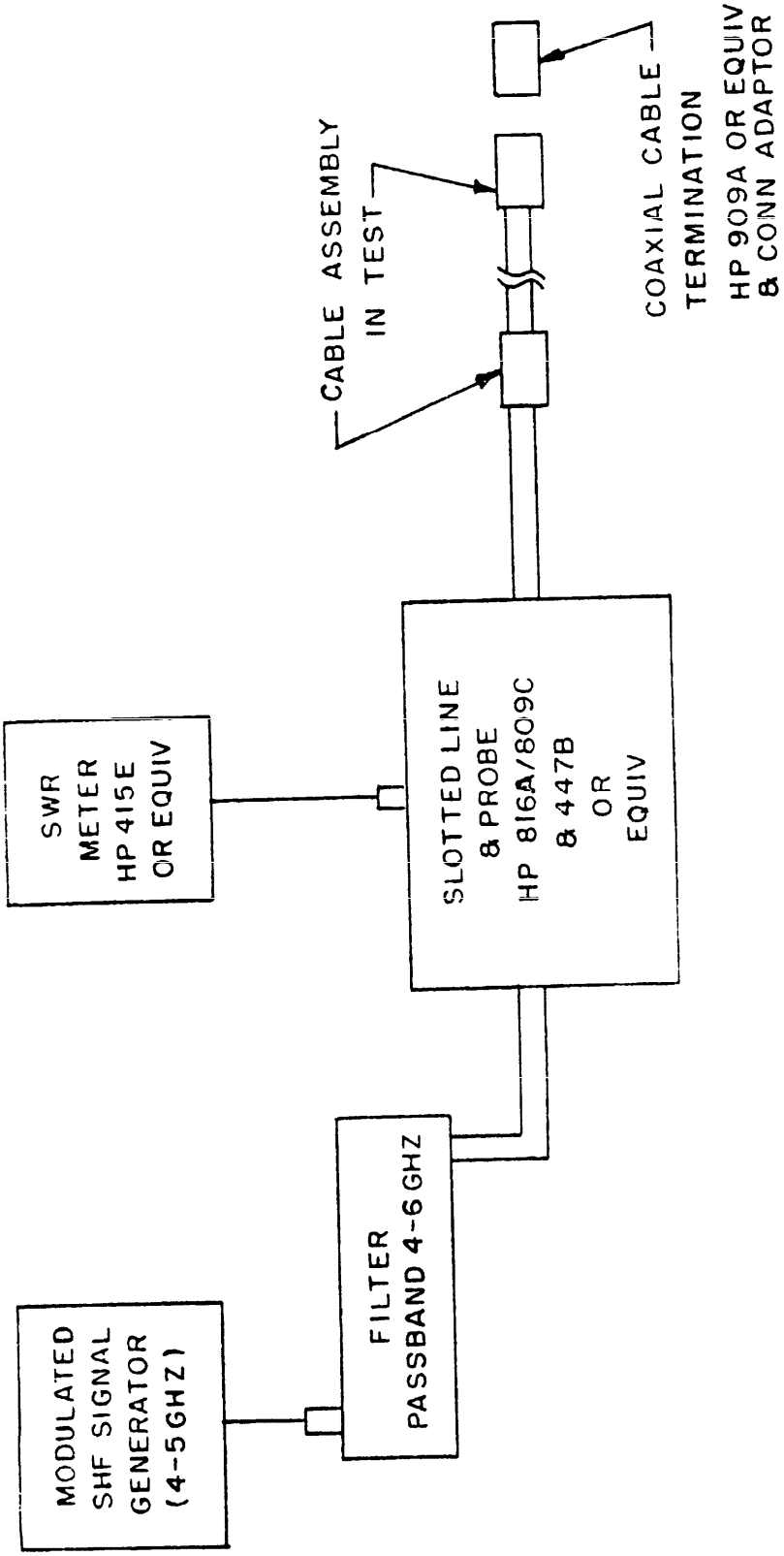
(f) Repeat 4.7.4.3b.

4.7.5 Waveguide Transmission Assemblies and Components Tests.- The tests of 4.7.5.1 and 4.7.5.2 shall be made on in-place receive and transmit waveguide systems. (Refer to Drawing No. SC-D-630303).

4.7.5.1 Voltage Standing Wave Ratio (VSWR).- The basic test set-up shall be as shown on Figure 3. Tuners and/or pads shall be included where necessary to reduce the residual VSWR of test set-up to a level which will assure accurate measurements. VSWR measurements shall be made as follows:

(a) Connect waveguide termination (HP G-910A or equal) to the TRANSMIT antenna waveguide port in antenna entrance box.

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FIG. 2 VSWR TEST SET-UP FOR RF COAXIAL CABLE ASSEMBLIES

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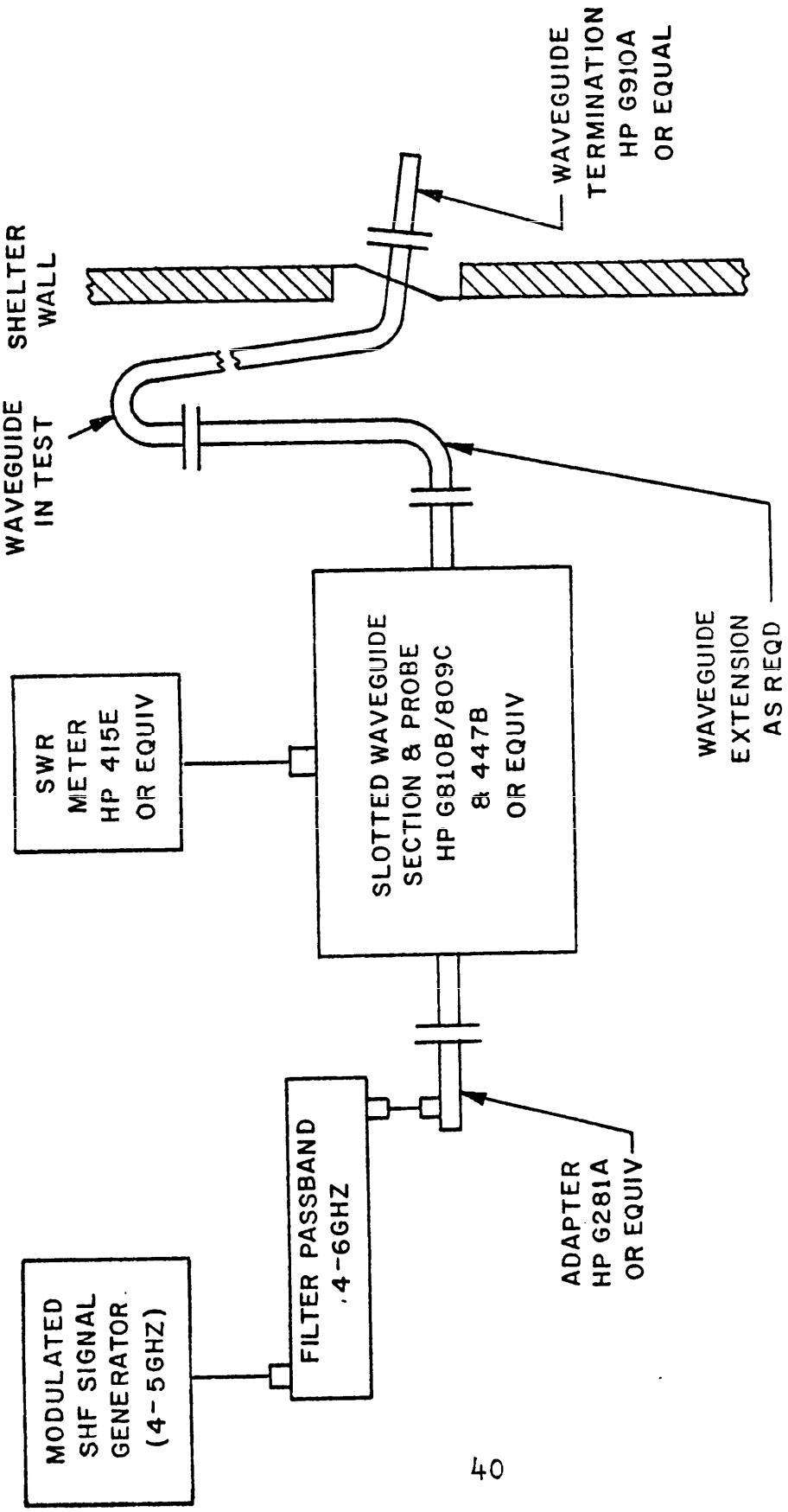


FIG 3. VSWR TEST SET-UP FOR WAVEGUIDE ASSEMBLIES

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(b) Place the facility waveguide switch in ANTENNA position.

(c) Connect slotted waveguide section (thru extension waveguide as required) to the AM-6090 power amplifier connecting waveguide flange.

(d) Measure VSWR at frequencies of 4.4 and 5.0 GH_z. The requirements of 3.3.5.1 shall be met.

(e) Place the facility waveguide switch in DUMMY load position.

(f) Repeat 4.7.5.1d.

(g) Connect waveguide termination to A RECEIVE antenna waveguide port in antenna entrance box.

(h) Connect slotted waveguide section to R-1287 Receiver A channel connecting waveguide flange.

(i) Repeat 4.7.5.1d.

(j) Repeat 4.7.5.1g and h on B receive waveguide assembly.

(k) Repeat 4.7.5.1d.

4.7.5.2 RF Power Input Test, Transmit Waveguide Assembly.- RF power input test shall be conducted as follows:

(a) Connect a high power (1,000 watts minimum) waveguide termination (dummy load) to the TRANSMIT antenna waveguide port in antenna entrance box.

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(b) Connect a 1,000 watt RF (4.4-5GHz) Power generator source, such as Power Amplifier, AM-6090 and suitable 150 MW exciter, to the facility AM-6090 Power Amplifier connecting waveguide flange.

(c) Place the facility waveguide switch in antenna position.

(d) Adjust generator for 1,000 watt output at frequency of 4.4GHz and apply power to the facility waveguide assembly for not less than 10 minutes. The requirements of 3.3.5.2 shall be met.

(e) Repeat 4.7.5.2d at frequency of 5.0 GHz.

(f) Place the facility waveguide switch in DUMMY LOAD position.

(g) Repeat 4.7.5.2d and e.

4.7.6 Control and Indicator Unit Operational Function Tests.- (See 3.3.6; reference drawing numbers SC-D-602485 and SC-D-630302).

(a) Test Preparation.-

(1) Fabricate test fixture shown on Figure 4 and connect fixture to facility control wiring terminations at connectors P6, P14 and P16 as indicated.

(2) Connect P20 and P21 to Control/indicator Unit.

(3) Connect a temporary jumper across the contacts of thermal switch, S11.

(4) Place AC power circuit breaker CB3 in OFF position.

(5) Place waveguide switch in the ANTENNA position.

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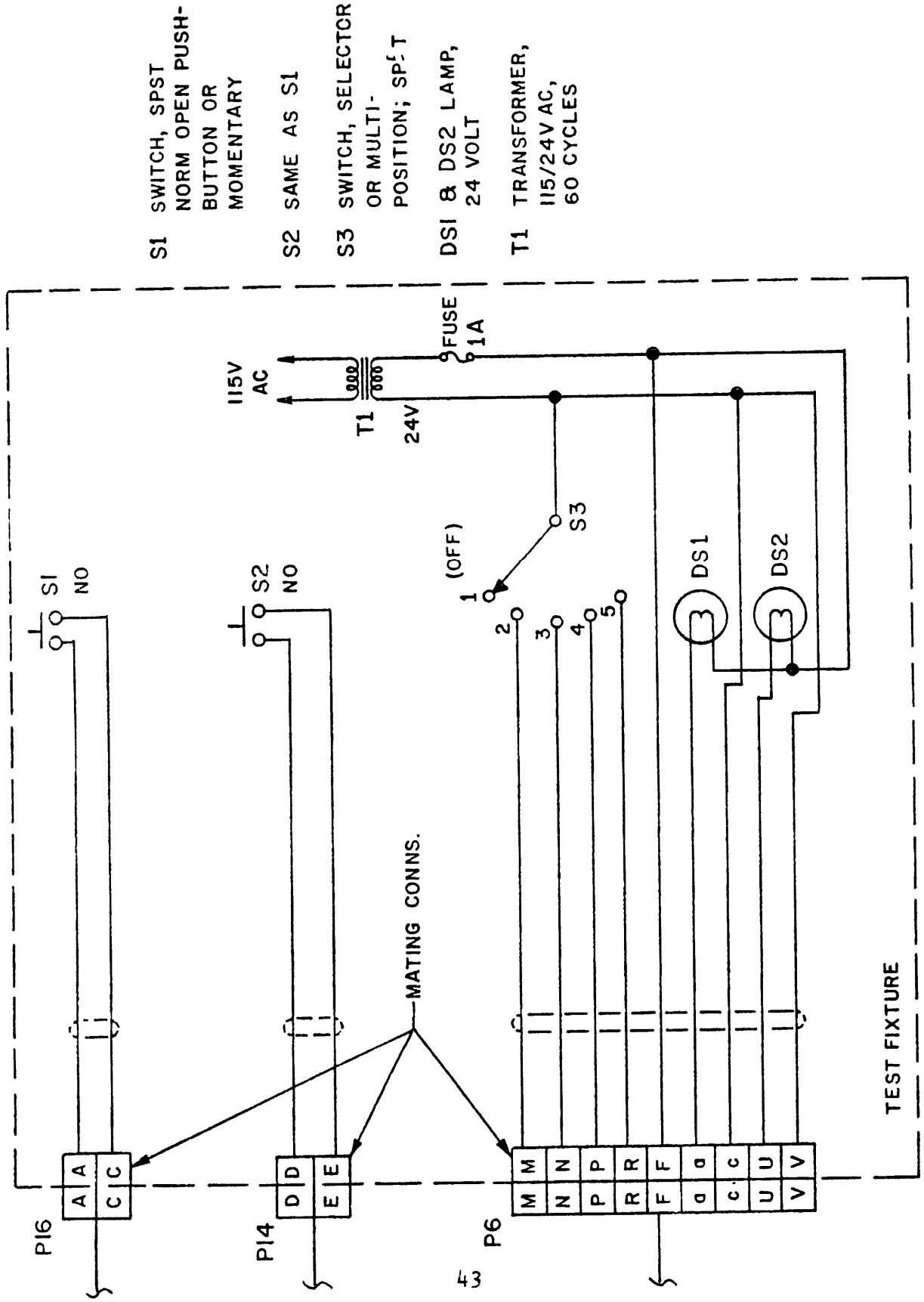


FIG 4. SIGNAL/CONTROL TEST FIXTURE

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(6) Set swtich S1 of Control/indicator Unit in ALARM disable position.

(7) Set switches S1 and S2 of test fixture in OPEN position.

(8) Set switch S3 of test fixture to position (OFF).

(9) Apply AC power to the test fixture.

(b) Test Procedure.-

(1) Apply 120/240 volts AC power to the Control/indicator Unit by placing circuit breaker, CB3 in the ON position. The following indicator lamps shall glow:

Green lamps B and C of DS-1 on Control/indicator Unit (pre-htr off).
Green lamps A and B of DS-4 on Control/indicator Unit (ant).
Red lamps A*, B*, C and D of DS-5 on Control/indicator Unit (aud alarm disabled).
Lamp DS-2 on test fixture.

(2) Depress button of switch, S1 on Control/indicator Unit. Red lamps A, B, C and D of DS-5 on Control/indicator Unit (aud alarm disabled) shall extinguish and lamp, DS-1 on test fixture shall glow.

(3) Position waveguide switch mid-way between antenna and dummy load switch positions. Lamp DS-2 on test fixture and green lamps A and B of DS-4 of Control/indicator Unit (ant) shall extinguish.

(4) Place waveguide switch in the DUMMY load position. Red lamps D and C of DS-4 on Control/indicator Unit (Dum Load) and lamp DS-2 on test fixture shall glow.

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(5) Rotate switch S3 on test fixture to Position 2. Red lamp A of DS-3 (traffic-in) on Control/indicator Unit shall glow.

(6) Rotate switch S3 on test fixture to Position 3. Green lamp D of DS-3 (traffic-in) on Control/indicator Unit shall glow.

(7) Rotate switch S3 on test fixture to Position 4. Red lamp B of DS-3 (traffic-out) on Control/indicator Unit shall glow.

(8) Rotate switch S3 on test fixture to Position 5. Green lamp C of DS-3 (traffic-out) on Control/indicator Unit shall glow.

(9) Close switch S2 on test fixture. Red lamps A B, C* and D of DS-2 on Control/indicator Unit (high-tem alert) shall glow.

(10) Close switch S1 on test fixture. Red lamps A and D* of DS-1 on Control/indicator Unit (pre-htr on) shall glow and power amplifier, AM-6090 air pre-heat elements shall get hot.

(11) With switch S1 on test fixture remainig closed, disconnect jumper across thermal switch, S11 (see 4.7.6a(3)). Red lamps A and D* of DS-1 on Control/indicator Unit (pre-htr on) shall extinguish; green lamps B and C of DS-1 (pre-htr off) shall glow; and, AM-6090 air pre-heat elements shall cool.

* Intermittent (flashing) glow

4.8 System Operational Test.- (See 3.4).

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4.8.1 Test Preparation.-

(a) Install all operational equipments in their respective locations. (See 6.1).

(b) Connect the shelter to an earth ground.

(c) Connect shelter facility to a 115/230 volt single phase AC power source.

(d) Establish test set-up as shown on Figure 5.

(e) Open all shelter air intake and exhaust ports.

(f) Set Power Amplifier, AM-6090 air intake and exhaust ducts for external air intake and discharge.

(g) Place waveguide switch in dummy load position.

(h) Place all AC power circuit breakers in ON position.

(i) Turn on shelter air exhaust blowers No. 1 and No.2.

(j) Place the 24 volt AN/GRC-106 main power switch, S9 in the DC converter position.

(k) Connect a 400 watt RF dummy load, consisting of a 4 ohm non-inductive resistor in series with a 90 MMF condenser, between the AN/GRC-106 antenna terminal (whip) and grounding terminal of radio set.

(l) Adjust Test Signal Unit of AN/GRC-143 for transmitter/receiver back-to-back operation.

(m) Set up Radio Set, AN/GRC-143 for 12-channel operation (¼ mile cable interface) by use of appropriate procedures contained on the charts in each cabinet.

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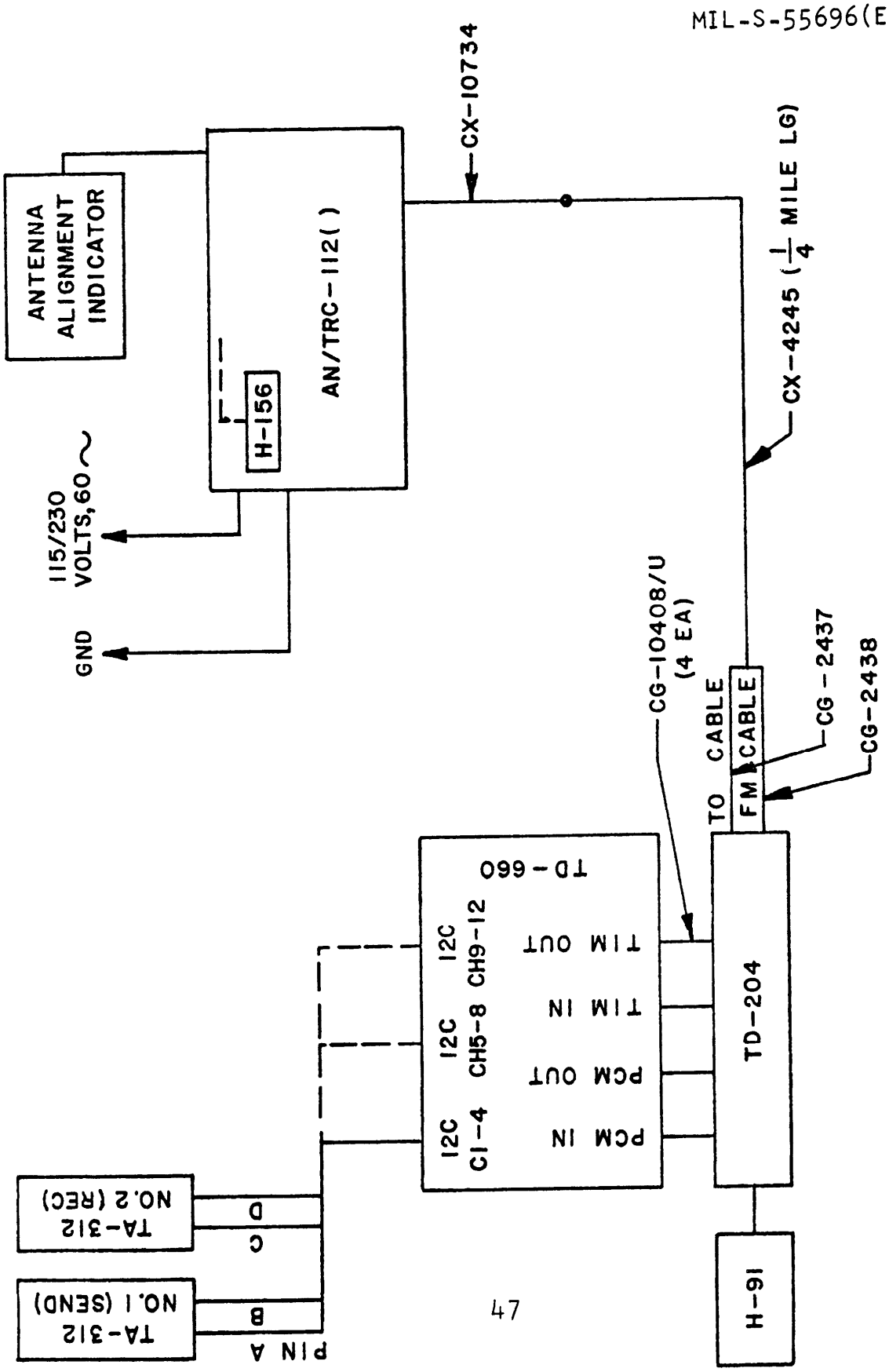


FIGURE 5. SYSTEM TEST SET-UP

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(n) Set up Radio Set AN/GRC-106 for CW operation on a frequency between 5.5 and 6.0 MHz.

(o) Set up test multiplexer, TD-660 for 12 channel, 4-wire, master operation. Place AUX switch in OUT position and set switches I, II, III and IV to OFF.

4.8.2 Test Procedure, AN/GRC-143().-

(a) Disconnect plug P7 from transmitter, T-96/GRC-106() connector J6; then depress the TALK-SIG switch on test multiplexer, TD-204. The "ring" light on the transmitter shall turn amber and loud speaker sound 1600 Hz tone.

(b) With plug P7 disconnected, depress the ring button on transmitter. The ring button shall turn amber and loudspeaker shall sound. On test multiplexer, TD-204, the CALL light shall light and buzzer shall sound. Reconnect plug P7 to T-96/GRC-143() connector J6.

(c) Place test multiplexer, TD-204 TALK-SIG switch in TALK position. Talk into microphone of head-set H-91 at the TD-204. The voice transmission shall be clearly understandable in the order wire hand-set (H-156) of AN/GRC-143.

(d) Depress the push-to-talk switch on AN/GRC-143 order wire hand-set (H-156) and talk into microphone. The voice transmission shall be clearly understandable in the head-set (H-91) at test multiplexer, TD-204.

(e) Talk into handset of test telephone set, TA-312 No. 1 (send). The voice transmission shall be clearly understandable in handset of TA-312 No. 2 (rec).

(f) Remove test telephone set connector from channel 1-4 receptacle of TD-660 and connect it to channel 5-8 receptacle. Repeat 4.8.2(e).

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(g) Remove test telephone set connector from channel 5-8 receptacle of TD-660 and connect it to channel 9-12 receptacle. Repeat 4.8.2(e).

(h) Place antenna alignment indicator switch in Rec A position. The meter shall indicate a reading other than zero. Turn off receiver A of R-1287. The meter shall indicate zero.

(i) Repeat 4.8.2(h) on Receiver B circuit of R-1287.

4.8.3 Test Procedure, AN/GRC-106.-

(a) Set power amplifier meter selector switch to PWR OUT.

(b) With AN/GRC-106 Radio Set adjusted for full output power, depress button on Handset and hold depressed for 2 minutes. Observe PA meter indication. The power amplifier meter shall indicate full output power and remain steady during the 2 minute period.

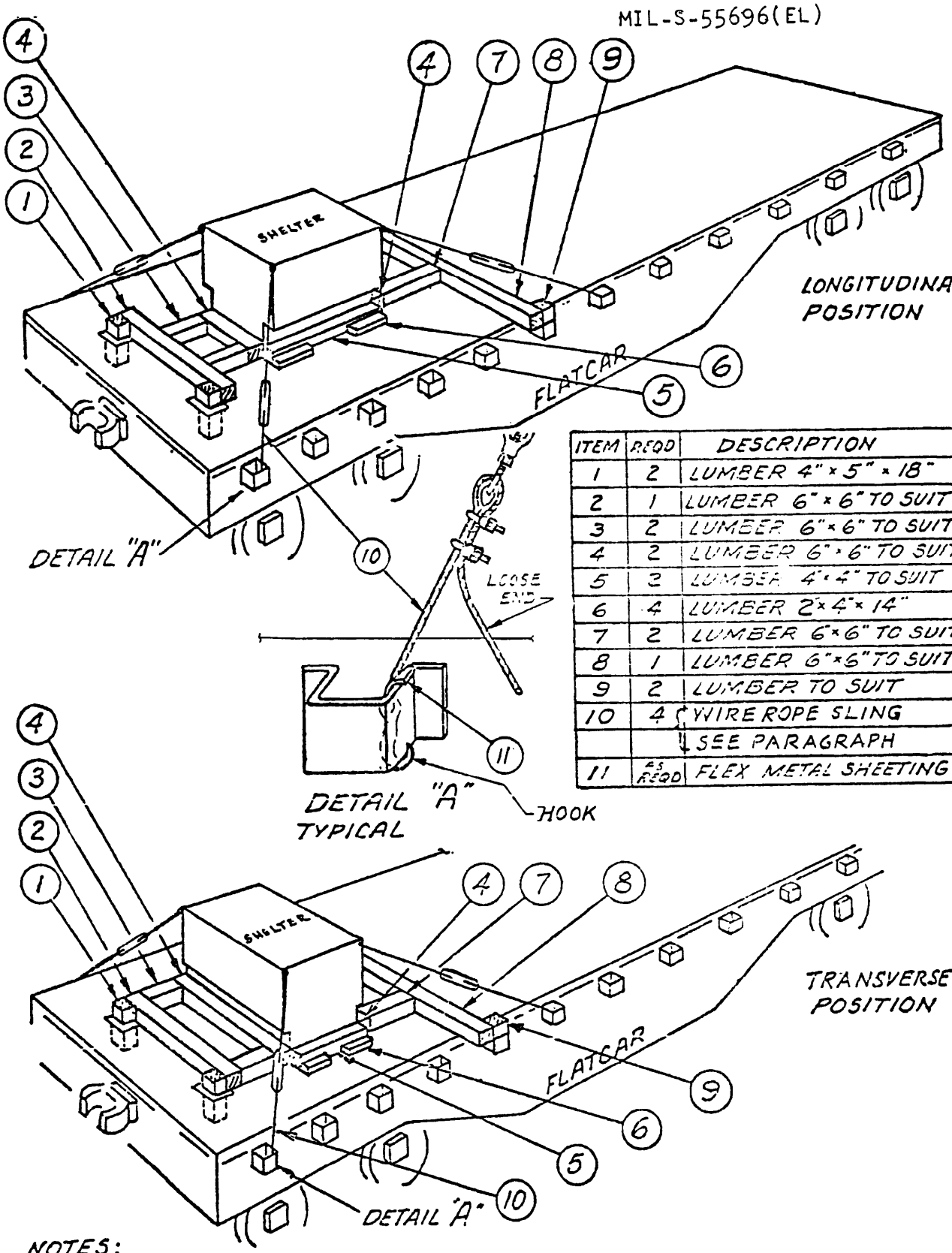
4.9 Vehicular transportation test.- The facility shall be secured (and blocked to prevent shifting) to a government furnished 1¼ ton, M-715, truck and transported over the specially prepared course described in 4.9.1 for 10 laps at a speed of 5 mph, 10 laps at a speed of 10 mph, 10 laps at a speed of 15 mph, and 10 laps at a speed of 18-20 mph. A lap is defined as traversing the course in one direction. The tires of truck shall be inflated to the cross-country driving pressure as indicated on the dashboard data plate. The shelter facility shall meet the requirements of paragraph 3.5 after completion of the test. Note: The facility shall be loaded to simulate the AN/TRC-112(V) using live loads (See 6.2b).

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4.9.1 Vehicular transportation course.- The course for the military truck transportation shall have twelve (12) 4-inch by 8-inch timbers placed 25 feet apart on the 8-inch face and with the 4-inch face fully above ground. The long top edges shall have 1 x 1-inch chamfers. The sixth and twelfth timbers shall be placed at 45 degrees to the direction of travel, and all other timbers shall be placed perpendicular to the direction of travel. The timbers shall be at least 16 feet long and the anchors shall be only near the ends of each timber. The timbers shall be anchored down securely by any convenient means. The ground on which the timbers are placed shall be hard and level.

4.10 Rail transport, impact test.- The shelter facility, with loads as specified in 4.9, shall be loaded onto a railroad flatcar. This test shall be performed on a level stretch of track in the manner described in this paragraph. The test flatcar, timber blocking, and shelter facility tiedown slings shall be as specified in Figure 6 and shall meet the provisions of the test as specified in 4.10.1; the test flatcar, less the shelter, and the two empty flatcars coupled to it shall each weigh 45,000 to 55,000 pounds. The loaded gondola car shall meet the provisions of the test as specified in 4.10.2 and shall have a total weight of 165,000 pounds. The loaded gondola car totaling 165,000 pounds traveling at 9 miles per hour (or equivalent energy conditions acceptable to the government) shall be impacted against the stationary loaded testcar coupled to the two empty flatcars, all with brakes off. Four (4) impacts shall be performed as follows: With the shelter facility positioned longitudinally on the flatcar, two impacts, one from each end of the shelter facility; with the shelter facility positioned transversely on the flatcar, two impacts, one from each side of the shelter facility. The lashing and blocking holding the shelter facility in place shall be tight at the start of each run. If the blocking is torn loose or the lashing is broken during the impact, the run shall be repeated. After each impact, the shelter facility shall meet the requirements of 3.6.

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ITEM	REQD	DESCRIPTION
1	2	LUMBER 4" × 5" × 18"
2	1	LUMBER 6" × 6" TO SUIT
3	2	LUMBER 6" × 6" TO SUIT
4	2	LUMBER 6" × 6" TO SUIT
5	2	LUMBER 4" × 4" TO SUIT
6	4	LUMBER 2" × 4" × 14"
7	2	LUMBER 6" × 6" TO SUIT
8	1	LUMBER 6" × 6" TO SUIT
9	2	LUMBER TO SUIT
10	4	WIRE ROPE SLING SEE PARAGRAPH
11	AS REQD	FLEX METAL SHEETING

NOTES:

1- THE BLOCKING LUMBER SHALL BE NOTCHED TO ACCOMODATE TOWING EYES, BOLTS AND RIVET HEADS.

2- THE BLOCKING LUMBER SHALL BE RIGIDLY FASTENED TO THE FLOOR OF THE FLATCAR.

FIGURE 6

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4.10.1 Test flatcar.- The wooden planking of the test flatcar bed shall be in good condition. All 6 x 6 inch blocking shall be secured to the bed of the flatcar using 10 inch spikes on 12 inch centers. The blocking along the door end and opposite end of the shelter shall abutt the corner castings along the full length of the castings except as necessary to clear the towing shackles; this shall be accomplished by means of power sawed notches. The blocking along the other two sides of the shelter shall abutt only the corner castings, along the full length of the castings. These four blocking members shall not be notched or shimmed to contact the trim along the sides of the shelter. The 4 x 4 inch and 2 x 4 inch blocking shall be nailed to the bed of the flatcar as shown in the figure referred to above. All other blocking shall be snugly abutting as shown in the figures. Lashing shall consist of four wire rope slings; each including 3/8 inch diameter, 6 x 19, flexible, steel, preformed-wire rope (having a breaking strength of 10,600 pounds minimum), a turnbuckle, two hooks, with thimbles and fasteners as required; the breaking strength of the sling shall be 9,000 pounds minimum. Before each impact, each sling shall be taut (600-800 pounds tension).

4.10.2 Loaded gondola car.- The load in the loaded gondola car shall be one of the following:

(a) A coarse size of either coal, crushed stone, gravel, crushed concrete, metal scrap (other than sheet metal or metal shavings or other springy forms of metal).

(b) Chunk, slab, block, or plate form of the preceding acceptable materials, constrained to prevent shifting during impact.

4.11 Drop tests.-

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4.11.1 Flat drop.- The shelter facility loaded as in 4.9 shall be lifted 18 inches from the ground and allowed to fall freely with the skids impacting onto a hard substantial concrete surface. The test shall be performed once. Upon completion of this drop, the facility shall meet the requirements of 3.7.

4.11.2 Rotational drop.- The shelter facility loaded as in 4.9 shall have one skid (or the knuckles of the skids) resting on a nominal 4 inch high timber. The outer edge of the opposite skid (or the opposite knuckles of the skids) shall be raised 18 inches and the shelter facility shall be allowed to fall freely onto a hard substantial concrete surface. This impact shall be performed four times, once for each bottom edge of the shelter facility. On completion of these drops, the shelter facility shall meet the requirements of 3.7.

4.12 Watertightness test.- The shelter facility, without the use of any additional external sealing, caulking, taping, etc. shall be subjected to the following two part test (Note: Perform tests with inside covers of entrance boxes removed).

(a) Each of the five exposed surfaces of the shelter facility, with covers closed, shall be sprayed with water from nozzles, model G29SQ (GG29SQ) as made by Spraying Systems Co., Bellwood, Illinois, or equal, pointing directly at the surface and arranged to provide uniform coverage of the surface. Each end panel shall be tested using nine nozzles, each side using nine nozzles, and the roof using four nozzles. Each nozzle shall operate at a pressure of forty psig at the nozzle, and shall be located nineteen inches from the panel with the exception of the roof nozzles which shall be at a distance of thirty-two inches. All of the five surfaces shall be simultaneously subjected to the test of this paragraph unless otherwise specified by the contracting officer. (See Figure 7 for nozzle location). Each panel shall be sprayed for a period

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of 20 minutes. (If the watertightness test is to be followed by the fording test of 4.13 the bottom row of nozzles may be eliminated with the exception of the nozzles facing the door-end panel). Upon completion of the above test, the shelter facility shall meet the requirements of 3.8. To determine that no water has entered the shelter panels, holes shall be made at points to be determined by the government at the time of testing. After this determination, the holes shall be sealed with rivets and sealer.

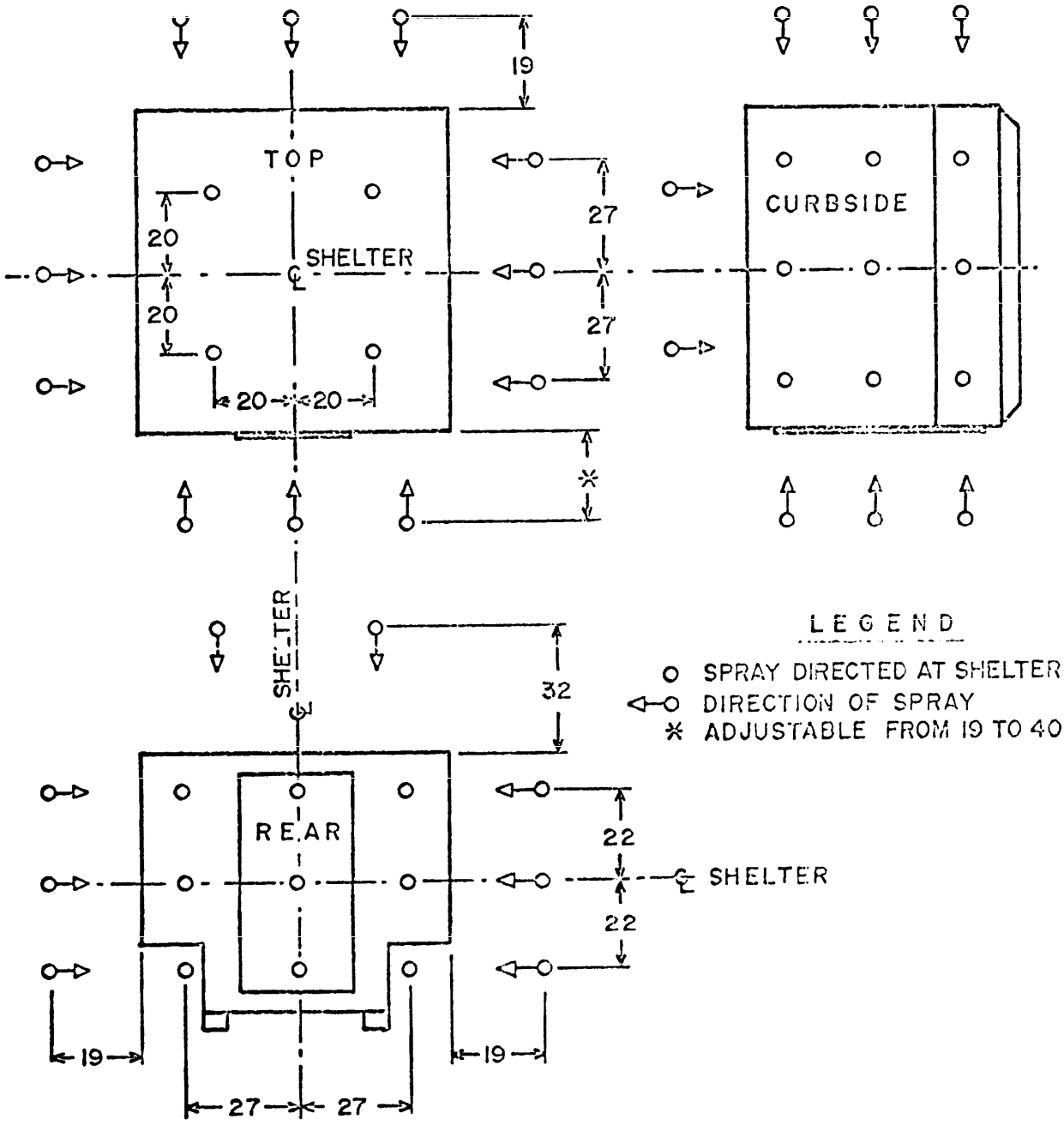
(b) Open all covers to operating position (with exception of the blower vent covers at the front of the shelter, the door vent panel at the rear of the shelter, AM-6090 intake and exhaust vent covers on roadside wall, and the air conditioning duct port covers on the curbside wall) and repeat a above.

4.13 Fording test.- The shelter facility shall be immersed in water for 1 hour to a depth of 26 inches (measured from the bottom of the skids) and meet the requirements of 3.9. No special fording kits shall be used. The shelter shall be restrained from floating by external tiedowns. To determine that no water has entered the shelter panels, holes shall be made at points to be determined by the government at the time of testing. After this determination, the holes shall be sealed with rivets and sealer.

4.14 Mounting insert tests.-

4.14.1 Preproduction insert test.- Four of each size insert shall be randomly selected for test. Each selected insert shall be tested with an axial load followed by a torque load in accordance with Table IV. After test, the requirements of 3.10 shall be met. Failure of any insert shall be cause for rejection of the sample shelter facility. This test shall be conducted prior to installation of brackets, racks, etc.

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NOTES

1 NOZZLES SHALL BE PERPENDICULAR TO SHELTER SURFACE.
2 UNLESS OTHERWISE SHOWN ALL DIMENSIONS SHALL BE TAKEN FROM INDICATED CENTER LINES.
3 GAGES SHALL BE PERMANENTLY INSTALLED IN PIPING TO INDICATE PRESSURE AT REMOTE NOZZLES.
4 ROADSIDE NOZZLE LOCATIONS SAME AS CURBSIDE.
5 FRONT NOZZLE LOCATIONS SAME AS REAR.
6 DIMENSIONS ARE IN INCHES.

NOZZLE LOCATIONS

FIGURE 7

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4.14.2 Production insert test.- Five of each size insert (if the total quantity of any size is less than five, all of that size shall be tested) shall be randomly selected for test in each sample shelter facility. Each selected insert shall be tested in accordance with Table IV. After test, the requirements of 3.10 shall be met. Failure of any insert shall be cause for rejection of the sample shelter facility. This test shall be conducted prior to installation of brackets, racks, etc.

Table IV - Mounting Inserts Loads

Insert size	Preproduction Test		Production Test	
	Axial(lbs)	Torque(in-lbs)	Axial(lbs)	Torque(in-lbs)
5/16-18	2000	100	1000	100
1/4-20	1300	60	650	60
#10-32	600	23	300	23

4.15 Testing of solderless terminal-lug connections.- Connections made with solderless terminal-lugs shall be tested for conformance to 3.20.4.

4.16 Inspection for interchangeability.- The dimensions listed below shall be gaged or measured to determine compliance with the physical interchangeability requirements of 3.21. When a dimension is not within specified or design limits, it shall be considered a defect.

(a) Size and mounting dimensions of brackets and mounting plates.

(b) Size and mounting dimensions of racks, patch panles and cabinets.

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4.17 Visual and mechanical inspection.- The shelter facility shall be examined for the defects listed in Standard MIL-STD-252, the requirements of 3.22 and Table V.

Table V - Classification of Visual & Mechanical Defects

Classification	Defects
Major	1. Mounting plates missing or mislocated
	2. Aircraft loading data plate, instruction or nameplate omitted, incorrect or illegible.
	3. Welds - cracked or porous.
	4. Sealer improperly applied (holes, separations, or lack of adhesion).
	5. Mounting inserts missing or mislocated.
	6. Burrs or sharp edges in wire duct not removed.
	7. Equipments or bracketry loose or missing.
	8. Doors, etc. inoperative.
	9. Electrical receptacles, switches, connectors, wire ducts or other electrical components loose, improperly located or inoperative.
	10. Locking or holding devices missing or inoperative.
	11. Gaps between sections of wire duct exceed 1/16".

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Table V - Classification of Visual & Mechanical Defects

Classification	Defects
Minor	1. Doors, not easily operable. 2. Fastening devices difficult to operate. 3. Finish - abrasions or scratches. 4. Burrs and sharp edges or brackets not removed. 5. Screws loose.

4.18 Quality conformance inspection of preparation for delivery.- Preparation for delivery shall be inspected in accordance with Specification MIL-P-116 to determine conformance to the requirements of Section 5.

4.19 Rough handling test (preparation for delivery).- When rough handling test in accordance with Specification MIL-P-116 is required by the contract, the following functional tests shall be conducted to determine freedom from operational malfunction caused by rough handling:

Paragraph 4.7.1	Power and Signal wiring
Paragraph 4.17	Visual and mechanical

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.- Preservation, packaging, and packing of the shelter facility shall be in accordance with applicable requirements of Specification MIL-S-55505.

6. NOTES

6.1 Intended use.- The shelter, Electrical Equipment S-336()/TRC-112 is the shelter facility portion of the AN/TRC-112() and it is used to house the operational equipments including:

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(a) 1 ea Radio Set, AN/GRC-143().

(b) 1 ea Radio Set, AN/GRC-106().

(c) 1 ea Power Supply, PP-4763/GRC.

(d) 1 ea Converter, Telegraph-Telephone Signal, CV-425()/U.

(e) 1 ea Telephone, TA-312()/PT.

(f) 1 ea Indicator, Antenna Alignment, ID-1708/GRC.

(g) 1 ea Handset, H-156/U.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification and any amendment thereto.

(b) Preproduction and Group C inspection of the shelter facility requires use of government furnished live loads (equipments listed in Para 6.1) unless otherwise specified in the bid request and contract.

(c) A list of government furnished equipments required as part of the S-336()/TRC-112 on order.

(d) A list of equipments to be government loaned for testing purposes.

(e) Submission of the statement referenced in 3.18.4 as soon as possible after award of contract. This statement should be submitted to the contracting officer.

6.3 Inspection.- Inspection is the examination or testing, or both, of supplies to determine compliance with applicable requirements. Sampling is an element of inspection.

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6.3.1 Examination.- Examination consists of simple, generally nondestructive determinations of compliance, without use of special testing equipment.

6.3.2 Testing.- Testing consists of determinations of compliance, using technical means.

6.4 Group C inspection.- Approval to ship may be withheld, at the discretion of the government, pending the decision from the contracting officer on the adequacy of corrective action. (See 4.5.3.3).

6.5 Nomenclature.- The parenthesis in the nomenclature will be deleted or replaced by a letter identifying the particular design, for example: S-336R/TRC-112. The contractor should apply for nomenclature in accordance with the applicable clause in the contract.

6.6 Verification inspection.- Verification by the government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

Custodian:
Army - EL

Preparing Activity:
Army - EL

Project No. 5820-A464

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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Fort Monmouth, New Jersey 07703

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